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CS62 Grading Contract– Roomer

Our Problem, and Intended Features

Roomer is a platform designed to make the housing process at Pomona College more accessible to its students. It will help students navigate the platform by making room information more transparent/easier to access, allow students to exchange or trade registration times, and provide a marketplace/valuation layer for students to make optimal housing-related decisions. Our current vision for the project includes 3 features: a room-information system, a draw-time exchange system, and an auction house w/ valuation tool.

Research, and Project Choice

We were motivated to create this project due to student frustration around housing drawing uncertainty, unequal information, and informal exchanges leading to selection advantages for some students. After interviewing rising sophomores and juniors regarding their plan for housing room draw, many expressed frustration and dissatisfaction after receiving a late room draw time since many of the "better" dorms would be taken by rising seniors. Some students thought about swapping room times with a friend of theirs who has a better time, but this is a process that we have identified as inefficient. The current process of swapping room times includes having the individual with the better time first securing the room for the individual with the not-so-good time, and then contacting the Director of Campus Life Operations who has to manually swap the rooms of the two individuals. Our implementation could improve the efficiency of this room swap process. Since a dorm building was removed for the 2026-2027 school year, these issues have been exacerbated, so we found it critical to take action now, so that we could create a potential solution by room draw next cycle.

Grading Contract

To pass the project (get a C), our group will complete the following:

- Submit a completed write-up with needfinding results and algorithmic/affordance analysis.
  - Explain how the project addresses a real-world issue, and hopefully solve the issue locally on our campus.
- Provide a csv file with between 10-999 unique data points (clarified with Prof. Li that this would not be individual rows).
- Submit a Github Repo including:

    - Our backend of the project.
    - Our frontend of the project.
    - A README.md file with a tutorial on how to compile and run the code, as well as interpreting the project (similar to a Hackathon deliverable).
- Use one data structure and create one working feature.
    - We will use a HashMap to create a Room Directory/ Room Search feature.
        - Data Structure: HashMap helps us map a unique room ID to each Room object, supports fast lookup, insertion, and updates (i.e. market status) and will be used as the main in-memory structure for the room directory.
        - Feature: Users will be able to browse rooms in the housing inventory, where each room stores information such as building, room number, occupancy, AC, and square footage. Users are able to retrieve a room by their ID, or filter for rooms by selecting certain attributes (ex: has AC).
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To get a B on the project, our group will complete the following:

- Complete everything required as detailed in "passing the project."
- Have our csv file include over 1000 entries.
- Our Github Repo will now include:
    - Documentation of the public API of our data structures.
- We will implement our 2nd feature, a Draw-Time exchange.
    - Feature: Users will be able to post their housing draw time for exchange. Users are able to view mutual exchange opportunities. Our system will be able to identify whether or not a direct trade is valid between two different users.
    - Data Structure: We will have two Hashmaps– one that stores draw slot information by unique ID, and one that maps user IDs to owned draw slots. It is possible we may also need to implement a graph to find direct matches/swap relationships to model potential exchanges for draw times.

To get an A on the project, our group will complete the following:

- Complete everything as listed above in "get[ting] a B."
- Our Github Repo will now include:
    - Detailed usage examples.
- Our code will include:
    - Good style, JavaDocs, and in-line comments.
- We will implement our 3rd feature, a Marketplace.

    - Feature: Users will be able to list their draw times for "sale" on the marketplace. Users can browse listings and get an estimated room cost/ determine the value of their draw-time. Users can submit bids/offers. Users can see a ranked market view.
        - We plan to implement this either as a list & offer market, or a valuation tool to estimate room desirability based on room attributes and historical selection data (or the combination of both).
    - Data Structure: We will have a HashMap to store listings by ID. We will also need to implement a balanced BST to sort listings or rooms by asking price or expected value by rank. We might need to implement a PriorityQueue if we want to implement an auction/bid system to store current offers for each listing, so the best offer can be returned easily. We will store our transactions in an ArrayList.

To get an A+ on the project, our group will complete the following:

- Complete everything as detailed in "get[ting] an A."
- Implement at least one of these extra features to create a front-end for the MVP.
    - Option 1: An Interactive Campus Map.
        - We will add a room & building based map interface so users are able to visually explore available housing by location, in the context of the 5C campus.
            - Users should be able to click on buildings to see rooms, and filter displayed rooms by occupancy, AC, value, etc.
    - Option 2: We will build a GUI for Roomer with Javascript/CSS (probably) as a frontend to connect to our Java backend.
        - Our GUI will allow users to browse rooms, search/filter inventory (available draw times to be sold), view/exchange opportunities, and browse marketplace listings.

Part 0 Slides

https://docs.google.com/presentation/d/1NaAQepK9rYbfz9uObo-OZx07rs0JSvIe8K67AzsNN5M/edit?usp=sharing

Part 1 Feedback

It would've been helpful to have a little bit longer for Part I/more flexible submission requirements (non-final dataset, non-final interfaces) because we've had to wait on getting all data from HRL, which has blocked us several times (we’re currently still waiting to hear back about dorm occupancy rates, unsure when this information will be available).